

The Coprime Cascade:

Structural Invariants, Effective Key Length, and Feedback Control in Rotor–Stepping-Switch Cipher Machines

Abstract

Electromechanical cipher machines were designed and defended in terms of the size of their key space, and they were broken in terms of something else. We give a structural account of what that something else is. We show that each of the two dominant wartime paradigms carries a proper algebraic invariant of its reachable permutation set — a reflector confines every encryption permutation to the conjugacy class of fixed-point-free involutions, a rotor bank freezes the signature of its stage permutation, and a split alphabet confines every permutation to the stabiliser of a partition — and that these invariants, not key-space size, are what the classical attacks consumed. We then prove that even a machine free of all three invariants is broken at the size of its *state* space rather than its key space, because static key material composed at the boundary of the cascade is transparent to the statistic that identifies a trajectory: we prove that the coincidence count of the reduced stream an attacker forms is independent of the plugboard, and that once a trajectory is fixed the plugboard is not searched but computed. This settles the effective key length of a cascade at $\log_2 |\Omega|$, where Ω is the reachable state set, and shows the plugboard’s 47.1 bits to be worth zero.

Guided by these two results we specify *Violet*, a cascade of a reflectorless rotor bank over $\mathbb{Z}_{26}$ and a Strowger switch bank over $\mathbb{Z}_{25}$ in which the rotor bank is driven irregularly by the inter-stage tap through key-set pin rings, while the switch bank remains an autonomous clock. We prove that the autonomous clock alone guarantees a period floor of 25^k letters under an arbitrary, adversarially chosen, message-dependent rotor rule, so feedback removes the counting structure the cryptanalyst needs without removing the period guarantee the operator needs. We give an exhaustive known-plaintext attack, prove its complexity is $\Theta(|\Omega|L)$, run it to completion on scaled instances up to 1.8×10^{11} states at 2.85×10^8 hypotheses per second, and find a unique surviving key after 10 to 18 letters of crib — the plugboard-consistency filter alone removes 2.68 bits per letter. Measured against the same instrumentation, the machine’s permutation family generates the full symmetric group S_{26} , its fixed-point count has mean 0.98 and variance 0.98 against the Poisson(1) prediction, a single changed letter alters 96.2% of the subsequent ciphertext against an ideal $25/26 = 96.15\%$, and two messages under one key share no state after four letters. Every structural result stated here — the period floor, the three invariant obstructions, the transparency theorem and correctness under an arbitrary stepping rule — is machine-checked in Lean 4 against *mathlib*, in a development of 45 theorems with no incomplete proofs.

1 Introduction

1.1 Two paradigms and a shared blind spot

By 1940 two families of cipher machine dominated. *Rotor machines* realise a time-varying substitution by passing a signal through a stack of wired discs whose relative displacement advances mechanically; the Enigma is the canonical example. *Stepping-switch machines* realise the same thing with banks of telephone-exchange selectors, each position of a bank wiring the

input terminals to the output terminals through an unrelated permutation; the Japanese Type B machine, called Purple by its American analysts, is the canonical example.

Both were broken, and the standard account of why is specific to each machine: the Enigma’s reflector guaranteed that no letter ever encrypted to itself, which turned every guessed crib into a filter and made Turing’s Bombe possible; the Type B machine split its alphabet into a group of six letters and a group of twenty, which let Friedman’s team attack two small problems instead of one large one. These explanations are correct. They are also, taken separately, an invitation to a bad inference — that a machine avoiding both defects is therefore secure. The natural experiment is to build such a machine: compose a reflectorless rotor bank with a stepping-switch bank over the whole alphabet, so that neither the reflector nor the split survives. That machine has a very large key space and no known classical entry point.

It is also broken at 2^{64} , and the reason is a property neither historical analysis names, because both machines have it and neither was broken through it first. We call it *open-loop transparency*: the state of the machine at time t is a function of t and the key alone. Depth attacks, crib dragging, Banburismus, the Bombe’s diagonal board and Purple’s analogue reconstruction are all, structurally, exploitations of this single fact. It survives the removal of the reflector, it survives the removal of the alphabet split, and it survives composition: a cascade of two open-loop stages is open-loop.

1.2 What this paper establishes

We make three contributions, in increasing order of how much they constrain design.

An invariant taxonomy. We show that the classical weaknesses are instances of one phenomenon: a proper algebraic invariant of the machine’s *reachable permutation set* $R \subseteq \text{Sym}(\mathcal{A})$. Enigma’s reflector forces R into the conjugacy class of fixed-point-free involutions, of size $25!! = 7.9 \times 10^{12} \approx 2^{42.8}$ out of $26! \approx 2^{88.4}$ (Theorem 9). A rotor bank forces its stage permutation into a single coset of the alternating group, because displacement is conjugation and conjugation cannot move a signature (Theorem 10); this is a one-bit leak available at every step, for every key, and it is a property of the rotor paradigm rather than of the Enigma. A split alphabet forces R into the stabiliser of a partition (Theorem 11). Each invariant is a free distinguisher, and each was the entry point of the attack that succeeded.

An effective-key theorem. We prove that in any cascade of the form $E_t = C_t \circ P$ with P static, the plugboard P is invisible to trajectory recovery in both the known-plaintext and the ciphertext-only setting. Under known plaintext it is not searched but read off, one letter per crib position (Theorem 20); under ciphertext only, the coincidence count of the reduced stream — the statistic that separates the correct trajectory from a wrong one — takes exactly the value it would take on the plaintext, for every P (Theorem 22). Therefore the work factor is $\log_2 |\Omega|$, the size of the reachable *state* set, and the 47.1 bits of plugboard in a ten-pair machine are worth nothing. We give the attack, prove its complexity, and run it.

A design and a period-floor theorem. The two results above are a specification: every bit of key must move the trajectory, and the trajectory must not be a function of time. Feedback achieves the second, and normally destroys the period guarantee that made autonomous stepping attractive in the first place. We show this trade is avoidable. In Violet the switch bank remains an autonomous base-25 odometer while the rotor bank steps under control of the inter-stage tap

through key-set pin rings, and we prove that the autonomous component alone forbids a state repetition inside 25^k letters *whatever the rotor rule does*, including under adversarial, message-dependent control (Theorem 19). The constraint that made the classical machines analysable is retained exactly where it produces a guarantee and removed exactly where it produces a vulnerability.

1.3 Scope

We prove an upper bound on security and exhibit a matching attack for the autonomous regime; for the tap-driven regime we prove the period floor, the diffusion behaviour and the destruction of depth, and we bound rather than determine the attacker's cost, because the pin rings admit a branch-and-prune search whose exact complexity we leave open (Section 12). We do not claim Violet is secure by modern standards — a 26-letter alphabet and a hand-set key cannot be — and Section 6 states precisely how many rotors and switches a given security level costs. What we do claim is that the accounting is now correct, that the design rules are forced rather than chosen, and that the central structural facts are machine-checked.

1.4 Organisation

Section 2 fixes notation. Section 3 proves the invariant taxonomy. Section 4 specifies the machine. Section 5 proves the period results. Section 6 proves the transparency theorem and the effective-key bound. Section 7 gives and measures the attack. Section 8 measures what feedback buys. Section 9 reports the statistical behaviour, Section 10 the mechanical realisability, and Section 11 the formal development. Section 12 lists what is open.

2 Preliminaries

Let $\mathcal{A} = \mathbb{Z}_n$ with $n = 26$, and let $\text{Sym}(\mathcal{A})$ be the symmetric group on $\mathcal{A}$, written multiplicatively with $(fg)(x) = f(g(x))$. Write $\tau \in \text{Sym}(\mathcal{A})$ for the cyclic shift $\tau(x) = x + 1$. For $f \in \text{Sym}(\mathcal{A})$, $\text{sgn}(f) \in \{\pm 1\}$ is its signature and $\text{Fix}(f) = \{x : f(x) = x\}$.

Definition 1 (Displacement). The *displacement* of a rotor $W \in \text{Sym}(\mathcal{A})$ by $s \in \mathbb{Z}_n$ is $W^{(s)} := \tau^{-s}W\tau^s$, i.e. $W^{(s)}(x) = W(x + s) - s$.

Physically, s is the angular position of the disc: rotating the disc relabels its contacts on both faces, which is conjugation by a shift. This one sentence is the source of Theorem 10.

Definition 2 (Rotor bank). A *rotor bank* of size r is a tuple $W_0, \dots, W_{r-1} \in \text{Sym}(\mathcal{A})$ with positions $p \in \mathbb{Z}_n^r$, realising

$$\rho(p) = W_{r-1}^{(p_{r-1})} \dots W_1^{(p_1)} W_0^{(p_0)}.$$

Definition 3 (Switch bank). A *stepping-switch bank* of size k over m levels is a tuple of *families* $U_j : \mathbb{Z}_m \rightarrow \text{Sym}(\mathcal{A})$, $0 \leq j < k$, with positions $q \in \mathbb{Z}_m^k$, realising $\sigma(q) = U_{k-1}(q_{k-1}) \dots U_0(q_0)$.

The asymmetry between the two definitions is the point of composing them. A rotor contributes n permutations that are all conjugate to one another; a switch contributes m permutations with no algebraic relation at all. A Strowger selector has $m = 25$ levels, and $\gcd(26, 25) = 1$.

Definition 4 (Plugboard). A *plugboard* with π pairs is an involution $P \in \text{Sym}(\mathcal{A})$ that is a product of π disjoint transpositions.

Definition 5 (Odometer). The *base- b odometer* on $\mathbb{Z}_b^d$ increments digit 0 and carries into digit $i + 1$ exactly when digit i wraps to 0.

Lemma 6 (Odometer linearisation). *A base- b odometer on d digits is the base- b digit representation of a counter in $\mathbb{Z}_{b^d}$: if $u^{(0)}$ has value v , then $u^{(t)}$ has value $v + t \bmod b^d$.*

Proof. Immediate by induction on t : the carry rule of the odometer is the carry rule of base- b addition of 1. $\square$

Lemma 6 is the reason a purely mechanical cipher is a counter in disguise, and it is used twice below with opposite intent: once to establish the exact period of the autonomous machine, and once to establish that its state is a function of time — which is what the attack of Section 7 consumes.

3 The invariant taxonomy

Definition 7 (Reachable set). For a machine M with key space $\mathcal{K}$ and state space Ω , the *reachable set* is $R(M) = \{E_{K,\omega} : K \in \mathcal{K}, \omega \in \Omega\} \subseteq \text{Sym}(\mathcal{A})$, the set of encryption permutations the machine can ever apply.

A cryptanalyst who knows $R(M) \subsetneq \text{Sym}(\mathcal{A})$ possesses a distinguisher that costs nothing and never expires. The three classical machines each hand one over.

3.1 The reflector obstruction

Definition 8. A *reflector* is an involution $U \in \text{Sym}(\mathcal{A})$ with $\text{Fix}(U) = \emptyset$.

Theorem 9 (Reflector obstruction). *Let U be a reflector and $g \in \text{Sym}(\mathcal{A})$ arbitrary. Then $E = gUg^{-1}$ satisfies $E^{-1} = E$ and $E(x) \neq x$ for all x . Consequently, for an Enigma-type machine, $R(M)$ is contained in the conjugacy class of fixed-point-free involutions, whose cardinality on 26 letters is $25!! = 7\,905\,853\,580\,625 \approx 2^{42.85}$, a fraction $2^{-45.5}$ of S_{26} .*

Proof. $E^2 = gUg^{-1}gUg^{-1} = gU^2g^{-1} = 1$, so $E^{-1} = E$. If $E(x) = x$ then $g(U(g^{-1}x)) = x$, hence $U(g^{-1}x) = g^{-1}x$, contradicting $\text{Fix}(U) = \emptyset$. The Enigma computes $E_t = S\Pi_t^{-1}U\Pi_t S$ with S the plugboard involution and Π_t the rotor composite, which is of the stated form with $g = S\Pi_t^{-1}$. The count of fixed-point-free involutions on 2ℓ points is $(2\ell - 1)!!$. $\square$

Two consequences are worth separating. The one exploited historically is $E(x) \neq x$: a guessed crib can be rejected as soon as it aligns a letter with itself, which happens with probability $1 - (25/26)^L$ on a random alignment of length L — 70% at $L = 32$. The one usually left implicit is quantitative: the machine can only ever apply $2^{42.8}$ of the $2^{88.4}$ permutations available to it, so more than half its nominal alphabet space is unreachable before any key is chosen. A corollary in the same breath: every Enigma permutation is a product of 13 transpositions and hence *odd*, and every Enigma permutation has cycle type 2^{13} — one cycle type, for all keys, for all time.

3.2 The signature obstruction

Theorem 10 (Signature invariance). *For any rotor bank $W_0, \dots, W_{r-1}$ and any positions $p \in \mathbb{Z}_n^r$,*

$$\text{sgn}(\rho(p)) = \prod_{i=0}^{r-1} \text{sgn}(W_i),$$

independently of p . A machine whose only moving stage is a rotor bank therefore has constant signature along its entire trajectory.

Proof. sgn is a homomorphism into the abelian group $\{\pm 1\}$, so $\text{sgn}(\tau^{-s}W\tau^s) = \text{sgn}(\tau)^{-s}\text{sgn}(W)\text{sgn}(\tau)^s = \text{sgn}(W)$, and the signature of a product is the product of signatures. $\square$

This is a one-bit distinguisher available for free at every step of every rotor machine ever built, and unlike Theorem 9 it does not depend on a reflector: removing the reflector does not remove it. It is also exactly the gap a stepping-switch bank fills. The levels $U_j(0), \dots, U_j(m-1)$ of a Strowger bank are unrelated permutations and generically carry both signatures, so $\text{sgn}(\sigma(q))$ varies with q and $\text{sgn}(E_t)$ with it. We measure 50.0% even permutations for Violet against 0% for Enigma (Figure 1b). This is the sharpest available answer to the question of what the switch paradigm contributes that the rotor paradigm provably cannot.

3.3 The partition obstruction

Theorem 11 (Partition obstruction). *Let $\emptyset \neq B \subsetneq \mathcal{A}$ and let $\text{Stab}(B) = \{f : f(B) = B\}$. Then $\text{Stab}(B)$ is a subgroup of $\text{Sym}(\mathcal{A})$ of order $|B|!|B^c|!$, and no $f \in \text{Stab}(B)$ maps a letter of B to a letter of B^c . For a Type B machine with $|B| = 6$, $|\text{Stab}(B)| = 6!20! \approx 2^{70.6}$, and the partition can be recovered from ciphertext alone by an orbit computation.*

Proof. Closure, identity and inverses are immediate; the map $f \mapsto (f|_B, f|_{B^c})$ is an isomorphism onto $\text{Sym}(B) \times \text{Sym}(B^c)$. The non-crossing claim is the definition. Recovery: the relation $x \sim y$ if some observed E_t maps x to y generates a partition refining $\{B, B^c\}$; with $O(n \log n)$ observed letters the two orbits separate. $\square$

The last clause is what makes the obstruction fatal rather than cosmetic. The Type B machine's input plugboard conjugates the internal partition into an unknown one, but conjugation preserves the *existence* of an invariant partition, and existence is what an orbit computation detects. Running the orbit computation on 4000 observed permutations recovers a smallest invariant block of size $6/26 = 0.231$ for Purple, and the trivial value 1.0 for Enigma, Violet and a uniform control (Figure 1c).

3.4 Escaping all three

Definition 12 (Invariant-free). A family $\{E_i\}$ is *invariant-free* if (i) some E_i has a fixed point, (ii) two members have opposite signature, and (iii) for every proper nonempty B , some E_i maps across B .

Condition (i) refutes the reflector class, (ii) refutes the signature coset and every rotor-bank representation, and (iii) refutes every partition. The three conditions are independent: Purple satisfies (i) and (ii) but not (iii); a reflectorless rotor machine satisfies (i) and (iii) but not (ii); Enigma satisfies (iii) alone.

Proposition 13 (Violet is invariant-free). *For the canonical build of Section 4, the reachable family satisfies Definition 12. Moreover the group generated by a sample of 58 reachable permutations is all of S_{26} .*

Proof (computational). Witnesses for (i) and (ii) are exhibited by direct evaluation; over 20 000 observed permutations the fixed-point count has mean 0.982 and variance 0.979 against the

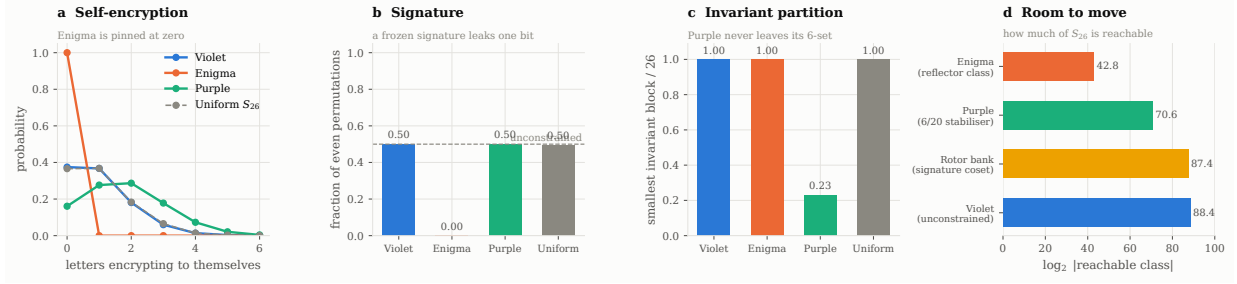


Figure 1. The three classical invariants, measured. (a) Enigma cannot encrypt a letter to itself; Violet’s fixed-point count is Poisson(1), matching a uniform draw from S_{26} . (b) A frozen signature is a free bit; the switch bank unfreezes it. (c) An orbit computation recovers Purple’s six-letter block from ciphertext alone. (d) How much of S_{26} each machine can reach before a key is chosen.

Poisson(1) prediction, and 50.0% are even. For (iii) the orbit computation of Theorem 11 returns the trivial partition. The group order is computed by Schreier–Sims and equals $26! = 4.03 \times 10^{26}$; membership of a generating set in S_{26} certifies that no invariant of any kind confines the reachable set to a proper subgroup. $\square$

4 The machine

4.1 Data path

Violet composes a rotor bank, a switch bank and a plugboard:

$$E_t = \sigma(q^{(t)}) \circ \rho(p^{(t)}) \circ P, \quad p^{(t)} \in \mathbb{Z}_{26}^r, \quad q^{(t)} \in \mathbb{Z}_{25}^k. \quad (1)$$

There is no reflector, so E_t is not an involution and $\text{Fix}(E_t)$ is generically nonempty; there is no alphabet split, so all 26 letters traverse every component. The *inter-stage tap* is the wire between the two banks,

$$g_t = \rho(p^{(t)})(P(x_t)) = \sigma(q^{(t)})^{-1}(y_t), \quad (2)$$

a physically existing conductor whose value the receiver can compute from the ciphertext letter and its own state before it needs to step.

4.2 Control path

Two regimes are defined on the same data path.

Definition 14 (Autonomous regime). $p^{(t+1)} = \text{odo}_{26}(p^{(t)})$ and $q^{(t+1)} = \text{odo}_{25}(q^{(t)})$.

Definition 15 (Tap-driven regime). The switch bank remains an autonomous odometer, $q^{(t+1)} = \text{odo}_{25}(q^{(t)})$. The rotor bank steps irregularly: rotor 0 always advances, and rotor $i \geq 1$ advances by $\text{pin}_i(g_t) \in \{0, 1\}$, where $\text{pin}_i : \mathbb{Z}_{26} \rightarrow \{0, 1\}$ is a key-set ring of 26 pins on rotor i .

The control rule is a Hagelin-style pin mechanism read by the tap conductor rather than by a lug cage, and the two paradigms are coupled in the control path rather than merely stacked in the data path: the switch bank sets the clock, the rotor bank obeys it irregularly, and the letter being enciphered decides how.

Proposition 16 (Correctness). *For every build, every key, every regime and every message x , decryption recovers x .*

Proof. By induction on message length. At each step the receiver holds the same state as the sender, computes $g_t = \sigma(q^{(t)})^{-1}(y_t)$ by (2), recovers $x_t = P^{-1}\rho(p^{(t)})^{-1}(g_t)$, and applies the same stepping rule to the same g_t , so the states agree at $t + 1$. The statement and proof are formalised for an arbitrary stepping rule in `Machine.decrypt.encrypt` (Section 11). $\square$

The proposition is stronger than it looks: it holds for *any* stepping rule whose argument is computable from the ciphertext and the current state. That is the precise design freedom feedback buys, and it costs nothing in synchronisation.

4.3 Parameters

We name two configurations.

	rotors r	chest	switches k	plug pairs	$\log_2 \Omega $	period
Violet 8/8 (field)	8	12	8	10	74.8	$650^8 = 3.19 \times 10^{22}$
Violet 14/14 (strategic)	14	18	14	10	130.8	$650^{14} = 2.40 \times 10^{39}$

5 Periodicity

Theorem 17 (Exact period, autonomous regime). *In the autonomous regime the joint state is the pair of counters $(v + t \bmod 26^r, w + t \bmod 25^k)$, and the state returns to its start exactly at multiples of $26^r 25^k$. The states at times $0, \dots, 26^r 25^k - 1$ are pairwise distinct.*

Proof. By Lemma 6 each bank is a counter, so the state at time t is $(v + t, w + t)$ in $\mathbb{Z}_{26^r} \times \mathbb{Z}_{25^k}$. It equals (v, w) iff $26^r \mid t$ and $25^k \mid t$. Since $\gcd(26, 25) = 1$ we have $\gcd(26^r, 25^k) = 1$, so the two divisibilities hold together iff $26^r 25^k \mid t$. Distinctness follows by applying this to $t' - t$. $\square$

Theorem 17 is the good news about coprimality and Theorem 22 will be the bad news about the same structure. It is worth stating the bad news in its own right here:

Corollary 18 (Open-loop transparency). *In the autonomous regime the state after t letters depends on t and the key only. In particular two messages of equal length encrypted under one key traverse identical state sequences, and a crib may be tested at position t without knowledge of the preceding plaintext.*

The tap-driven regime destroys Corollary 18 by construction. The concern this normally raises is that a message-dependent stepping rule carries no period guarantee: nothing prevents an unlucky message from driving the machine into a short cycle. Retaining one autonomous component answers the concern completely.

Theorem 19 (Period floor under arbitrary control). *Let the state be $(p, q) \in \Pi \times \mathbb{Z}_K$ for an arbitrary set Π , and let the trajectory be*

$$p^{(t+1)} = F_t(p^{(t)}, q^{(t)}), \quad q^{(t+1)} = q^{(t)} + 1,$$

where F_t is arbitrary — it may depend on the time, the plaintext, the ciphertext, the pin rings, and the whole history. Then the states at times $0, \dots, K - 1$ are pairwise distinct. For Violet, $K = 25^k$, giving a floor of $25^{14} > 2^{65}$ letters in the strategic configuration.

Proof. Project onto the second coordinate: $q^{(t)} = q^{(0)} + t$ in $\mathbb{Z}_K$. If $t, t' < K$ and the states agree then $t \equiv t' \pmod{K}$, so $t = t'$. $\square$

The proof is three lines and the design consequence is the paper’s central one. Feedback and a period guarantee are usually presented as a trade; they are not, provided one component of the state remains autonomous and its modulus is the guarantee. The bottleneck that made the classical machines analysable — a clock that ticks regardless of the message — is retained exactly where it produces a theorem and removed exactly where it produced the vulnerability.

6 Security accounting

6.1 The transparency theorem

Write the cascade as $E_t = C_t \circ P$, where $C_t = \sigma(q^{(t)})\rho(p^{(t)})$ is the moving composite and P is static. An attacker who hypothesises a trajectory $\hat{C}$ forms the *reduced stream* $z_t = \hat{C}_t^{-1}(y_t)$.

Theorem 20 (Plugboard determination). *If $y_t = C_t(P(x_t))$ for all t , then $P(x_t) = C_t^{-1}(y_t)$. Hence under the correct trajectory hypothesis each known-plaintext position determines one value of P outright, and P is recovered in time $O(L)$ once the trajectory is known. If the crib exercises every letter, P is unique.*

Proof. Apply C_t^{-1} to both sides. Uniqueness: two plugboards agreeing on a surjective x agree everywhere. $\square$

Corollary 21 (Repeated-letter test). *If $x_s = x_t$ then $C_s^{-1}(y_s) = C_t^{-1}(y_t)$, an equation in the trajectory alone. A crib therefore yields plugboard-free consistency tests at a rate governed by the letter-repetition density of the language.*

Corollary 21 is the structural analogue of Turing’s loop test, obtained here without a reflector and without a diagonal board: it is the one-sided plugboard, not the reflector, that makes it available.

Theorem 22 (Static-layer transparency). *Let $\text{Co}_L(z) = \#\{(i, j) : i < j < L, z_i = z_j\}$ be the coincidence count. Then $\text{Co}_L(P \circ x) = \text{Co}_L(x)$ for every permutation P . Consequently, under the correct trajectory hypothesis the coincidence count of the reduced stream equals that of the plaintext, for every plugboard; the statistic that identifies the trajectory is independent of P .*

Proof. P is injective, so $P(x_i) = P(x_j) \iff x_i = x_j$ and the two counted sets are equal. $\square$

Corollary 23 (Effective key length). *For a cascade with public wiring, known-plaintext key recovery costs $\Theta(|\Omega|L)$ where Ω is the reachable state set, independently of the plugboard’s size. In particular the plugboard’s $\log_2 \binom{26}{2, \dots} = 47.1$ bits contribute 0 bits of work factor.*

We stress what is and is not proved. Corollary 23 is an *upper* bound on security — an attack achieving it is given in Section 7. We do not prove a matching lower bound; no attack below $|\Omega|$ is known for the autonomous cascade, and the gap is stated as open in Section 12.

Design rule 24 (Key material must move). Every bit of key must alter the state trajectory. Key material composed at the boundary of a cascade is inert: it neither changes the trajectory nor perturbs any relabelling-invariant statistic of the reduced stream.

Design rule 24 is what puts the pin rings in Definition 15. Pins are not an ornament on the plugboard’s model; they are the plugboard’s key material moved from the boundary, where Theorem 22 makes it worthless, into the stepping law, where it is not.

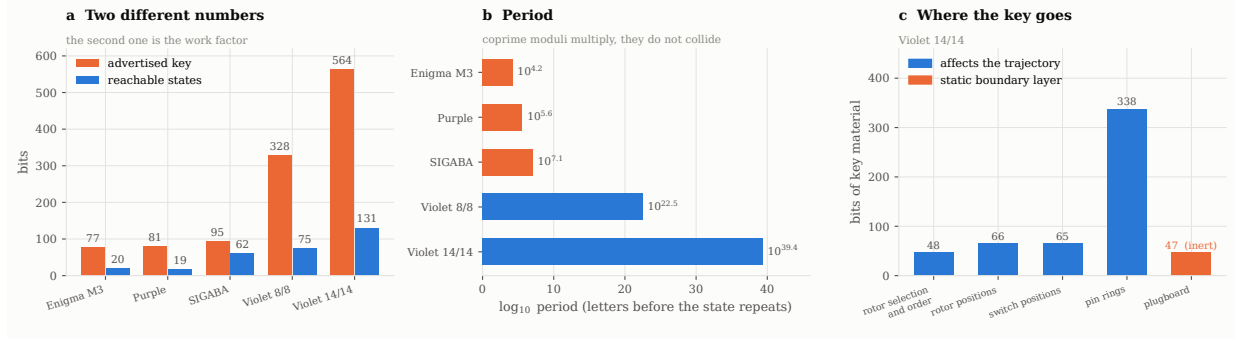


Figure 2. Security accounting. (a) Advertised key length against the size of the reachable state set, which is what the attack of Section 7 enumerates. (b) Period, on a log scale; coprime moduli multiply. (c) Where the key of a Violet 14/14 goes: everything that moves the trajectory counts, the boundary layer does not.

6.2 How much hardware a security level costs

Corollary 23 converts directly into a specification. Ignoring pin rings, whose contribution we bound rather than determine, a cascade attains s bits of proven-floor security iff

$$r \log_2 26 + k \log_2 25 \geq s. \quad (3)$$

With $r = k$ this is $r \geq s/9.31$: $s = 128$ needs $r = k = 14$. The corresponding unicity distance, at an English redundancy of $\log_2 26 - 1.5 = 3.20$ bits per letter, is $130.8/3.20 = 40.9$ letters — about eight words of crib is all the information the cipher can hide behind.

7 Key recovery

7.1 The algorithm

Input: public build; crib $x_{0..L-1}$ with ciphertext $y_{0..L-1}$.

For each state hypothesis $(c, d) \in \mathbb{Z}_{26^r} \times \mathbb{Z}_{25^k}$:

1. For $t = 0, 1, \dots$: set $z_t = \rho_{c+t}^{-1}(\sigma_{d+t}^{-1}(y_t))$.
2. Maintain the partial map $x_t \mapsto z_t$; reject as soon as it fails to extend to an involution with at most π transpositions and at most $26 - 2\pi$ fixed points.

Output: the surviving (c, d) , and P read off by Theorem 20.

The plugboard is never enumerated: it appears only as a consistency predicate, and the predicate is checked incrementally, so a wrong hypothesis is killed after a constant number of letters in expectation.

Theorem 25 (Attack complexity). *The algorithm terminates in time $O(|\Omega| \cdot \bar{L})$ where $\bar{L} = O(1)$ is the expected rejection depth, and uses $O(26^r + 25^k)$ memory for the stage tables. It outputs the true key, and the expected number of spurious survivors after L crib letters is $|\Omega| \cdot \Pr[\text{random map passes}]$, which falls below 1 at $L \approx \log_2 |\Omega| / D_{\text{filt}}$, where $D_{\text{filt}} \leq D$ is the number of bits the consistency test removes per letter and D is the redundancy of the plaintext language.*

Proof. Correctness is Theorem 20: the true hypothesis yields $z_t = P(x_t)$, which is consistent by construction. The cost statement follows because each hypothesis is tested by table lookups, and the survival probability of a wrong hypothesis decays geometrically in t , so the expected work per hypothesis is constant. The survivor estimate is the standard unicity-distance argument applied with key entropy $\log_2 |\Omega|$. $\square$

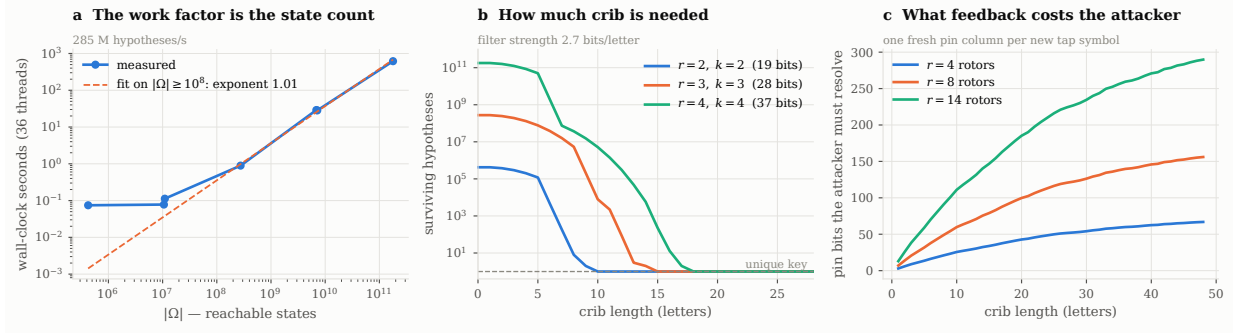


Figure 3. Key recovery, measured. (a) Exhaustive trajectory search over seven scaled builds; the fitted exponent confirms cost $\Theta(|\Omega|)$. (b) Surviving hypotheses against crib length; a unique key emerges at the unicity distance, with the plugboard never enumerated. (c) Under tap-driven stepping the attacker must additionally resolve one column of the pin matrix per distinct tap symbol.

7.2 Measurements

We instantiated the attack in C with OpenMP and ran it to completion on seven scaled builds, from $|\Omega| = 4.2 \times 10^5$ to $|\Omega| = 1.8 \times 10^{11}$, on 36 cores. In every run exactly one hypothesis survived, and it was the true one. Figure 3a shows wall-clock time against $|\Omega|$; the fitted exponent on the runs large enough for table construction to be negligible is 1.01, at a sustained 2.85×10^8 hypotheses per second, confirming the linear scaling of Theorem 25. Figure 3b shows the survivor count against crib length. The plugboard-consistency filter removes 2.68 bits per letter — below the 3.20 bits of English redundancy, as it must be, since the filter uses no language model at all — and a unique key emerges after 10 letters at 18.7 key bits and 18 letters at 37.4. The observed unicity distance therefore tracks $\log_2 |\Omega|/2.68$ rather than the information-theoretic $\log_2 |\Omega|/3.20$; a filter that also scored English statistics would close the difference.

Extrapolating the measured throughput to the canonical builds, Violet 8/8 falls at $2^{74.8}$ and Violet 14/14 at $2^{130.8}$ state hypotheses — the first within reach of a determined adversary with modern hardware, the second not. The distance between the two is eight bits of rotor and eight of switch, and nothing else in the key matters to it.

7.3 The attack against the tap-driven regime

The algorithm above does not apply to Definition 15, because step 1 presumes $p^{(t)} = c + t$. An attacker must instead guess the pin columns as they are first used. Each newly seen tap value exposes one fresh column of the $(r-1) \times 26$ pin matrix, so after L crib letters the pin entropy in play is $(r-1) \cdot d(L)$ bits, where $d(L)$ is the number of distinct tap symbols observed; d saturates at 26. Figure 3c plots the measured $(r-1)d(L)$. Against this the crib supplies about $\log_2 26 = 4.7$ bits of constraint per letter, so a branch-and-prune search is a race between a branching factor of 2^{r-1} on fresh symbols and a pruning factor of $1/26$ per letter. For $r \leq 5$ pruning wins and the search is feasible; for $r = 14$ the branching factor is 2^{13} against a pruning factor of $2^{-4.7}$ and the tree grows. We do not claim a bound here; determining the true complexity of pin recovery is the main open problem left by this work.

8 What feedback buys

Theorem 26 (Error propagation). *Model E_t at distinct states as independent uniform draws from $\text{Sym}(\mathcal{A})$. In the tap-driven regime, if two messages first differ at position i , then for every $j > i$ the ciphertexts differ at j with probability $1 - 1/26$, up to a resynchronisation term bounded by $j/|\Omega|$. In the autonomous regime the probability is 0 for all $j > i$.*

Proof. Autonomous: the states agree at every j , so y_j depends only on x_j . Tap-driven: at position i the taps differ, so the rotor increments differ with probability $1 - 2^{-(r-1)}$ over the pin ring, and once the states differ the two permutations are independent draws, giving agreement probability $1/26$ per position. Resynchronisation requires the two rotor states to coincide again while the switch states already agree, an event of probability at most $1/26^r$ per step by Theorem 19; union bound over j steps. $\square$

Measured over 60 keys and 200 positions the tap-driven change rate is 0.9624 against the predicted $25/26 = 0.9615$, and the autonomous rate is exactly 0 (Figure 4a,c). For the strategic configuration the resynchronisation bound is 1.6×10^{-20} per step.

Theorem 27 (Destruction of depth). *In the autonomous regime two messages of equal length under one key induce identical state sequences. In the tap-driven regime the sequences separate at the first position where the taps differ and, up to the resynchronisation term of Theorem 26, do not rejoin.*

Measured over 40 key/message pairs, autonomous trajectories agree at every one of 120 positions and tap-driven trajectories agree at none after position four (Figure 4b). Depth is the resource that Banburismus, the Bombe’s menu construction and Purple’s reconstruction all consumed; the tap-driven regime does not supply it.

The cost is the standard cost of self-synchronising feedback: a garbled ciphertext letter corrupts the following letters until the state resynchronises, which for the tap-driven rule it does not do. Operationally, messages must be short enough that a transmission error is cheaper to retransmit than to reconstruct — the same discipline that applies to any chained cipher, and the reason the field configuration exists.

9 Statistical behaviour

Over a 20 000-letter English message the ciphertext χ^2 against uniform falls from 12 472 to 20.3 on 25 degrees of freedom (critical value 37.65 at $\alpha = 0.05$), and the index of coincidence falls from 0.0624 to 0.03845 against the uniform value $1/26 = 0.038462$ (Figure 5a,b). Both regimes give the same answer to three digits, which is the expected result: flattening is a property of the data path, and the two regimes share it.

The third panel is the one that matters. For 40 independent plugboards we computed the coincidence index of the reduced stream under the correct trajectory and under 240 wrong ones. The correct trajectory returns the plaintext value 0.0653; wrong trajectories return 0.0384; the separation is 61.4 standard deviations and does not vary with the plugboard, exactly as Theorem 22 requires. An attacker sorting state hypotheses by this statistic never needs to consider P at all.

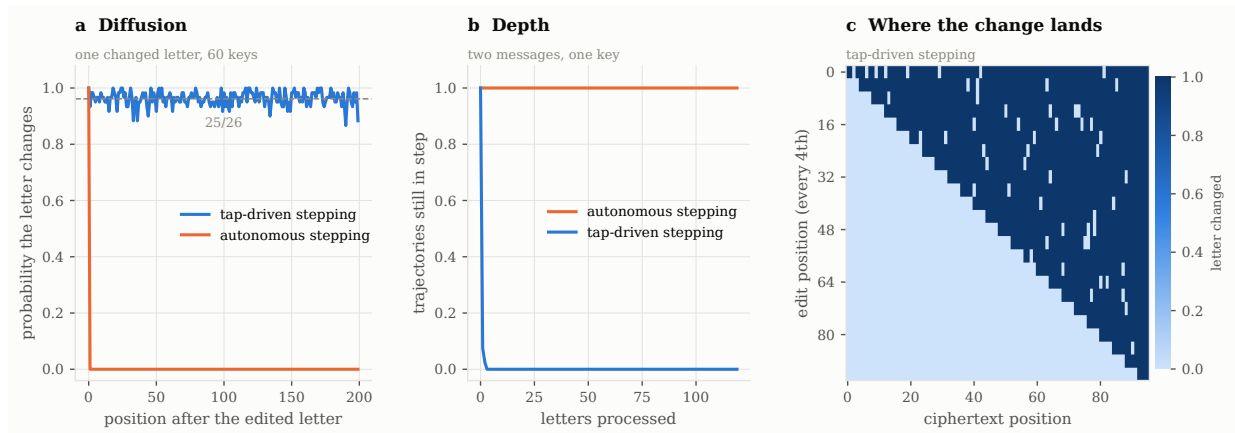


Figure 4. Feedback in the control path. (a) A single changed letter alters 96.2% of all later ciphertext under tap-driven stepping and none of it under autonomous stepping. (b) Two messages under one key: autonomous trajectories stay locked, tap-driven ones separate within four letters. (c) The same edit applied at 24 positions.

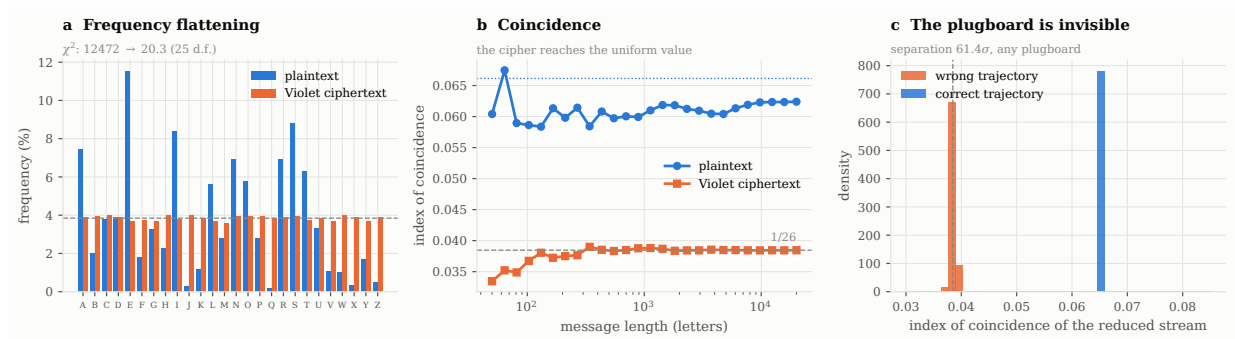


Figure 5. Ciphertext statistics. (a) English letter frequencies against Violet ciphertext. (b) Index of coincidence against message length. (c) The statistic that identifies a trajectory, computed under 40 different plugboards: the separation is 61.4σ and the plugboard is invisible to it.

10 Mechanical realisability

Every component is period technology. The rotor bank is a stack of wired discs with a Geneva-type advance, as in the Enigma and the SIGABA; a 14-rotor stack is heavier than the SIGABA’s 15 but of the same order. The switch bank is 14 Strowger selectors, each a 25-level unisector of the kind installed by the hundred in a telephone exchange. The plugboard is a patch panel. The pin rings are Hagelin’s mechanism: a 26-pin ring per rotor, each pin set left or right by hand, read by a follower.

The only element without an exact wartime precedent is the tap reader: a decoder that converts the 26-way inter-stage conductor into the $r - 1$ step/no-step signals for rotors $1..r - 1$. It is a 26-input, $(r - 1)$ -output diode or relay matrix whose $26 \times (r - 1)$ crosspoints are the pin settings, energised from the conductor that is already carrying the signal. In 1943 terms this is a relay tree of the sort used for translator circuits in exchanges, and it adds one relay step to the per-character cycle. The key-setting burden is r rotor positions, k switch positions, π plug pairs and $(r - 1) \times 26$ pins; the pins are a long-period key, changed weekly, exactly as Hagelin’s were.

11 Formal development

The structural results are machine-checked in Lean 4 against *mathlib*: a development of 45 theorems across five files, compiling with no incomplete proofs and no axioms beyond *mathlib*'s.

Result	Lean name	Content
Prop. 16	<code>Machine.decrypt_encrypt</code>	Decryption inverts encryption for an <i>arbitrary</i> stepping rule
—	<code>Machine.encrypt_injective</code>	The map is a cipher
Cor. 18	<code>Machine.runState_autonomous</code>	Autonomous state is a function of message <i>length</i>
—	<code>Machine.autonomous_depth</code>	Equal-length messages share a trajectory
Thm. 17	<code>jointStep_fixed_iff</code>	Return time is exactly lcm, via coprimality
—	<code>jointStep_injOn</code>	No repeat inside one period
Thm. 19	<code>closed_loop_period_floor</code>	Floor K under an arbitrary rotor rule
Thm. 9	<code>IsReflector.conj</code>	Conjugates of a reflector are reflectors
—	<code>reflector_no_self_encryption</code>	$E(x) \neq x$
Thm. 10	<code>sign_rotorStage</code>	$\text{sgn}\rho(p) = \prod_i \text{sgn}W_i$, for all p
Thm. 11	<code>blockStabilizer</code>	Block-respecting permutations form a subgroup
Thm. 20	<code>plug_determined, plug_unique</code>	Plugboard computed, and unique
Cor. 21	<code>repeat_letter_test</code>	Plugboard-free equations from repeats
Thm. 22	<code>coincidences_reduce_eq</code>	Coincidence count independent of P
Def. 12	<code>InvariantFree</code>	The three escape conditions and their consequences

Two choices in the formalisation are worth noting. First, `Machine.decrypt_encrypt` is stated for a machine whose stepping rule consumes an arbitrary symbol computable by the receiver, so a single proof covers both regimes and any future rule of the same shape. Second, `closed_loop_period_floor` quantifies over an arbitrary time-dependent rotor rule F , which is the strongest form of the statement: it grants the adversary complete control of the rotor bank and still forbids a repeat.

12 Open problems

1. **A lower bound for the autonomous cascade.** We prove security is at most $\log_2 |\Omega|$ and know no attack below it. Is Ω -enumeration optimal, or does the coprime structure admit a meet-in-the-middle below $|\Omega|$? The differential at period offsets is suggestive: $E_{t+25^k}^{-1} E_t$ eliminates the switch bank and the plugboard simultaneously.
2. **The complexity of pin recovery.** Section 7 identifies a branch-and-prune search with branching 2^{r-1} on fresh tap symbols against $\log_2 26$ bits of constraint per letter. The threshold in r at which pruning stops winning is not determined here.
3. **Optimal control couplings.** Definition 15 taps the inter-stage wire. Other taps — the switch-bank output, a second rotor sub-bank, a SIGABA-style index layer — give different trade-offs between diffusion rate and pin entropy. Which coupling maximises the proven floor per unit of hardware is open.
4. **Formalising the attack.** The transparency theorems are verified; the complexity statement of Theorem 25 is not, because a cost model for the search would have to be formalised first.

13 Related work

Rejewski’s recovery of the Enigma wiring from permutation cycle structure [8] and Turing’s crib-based method [14] are the origin of the analysis in Section 7; the repeated-letter test of Corollary 21 plays the role Turing’s loops play, and needs no reflector. Deavours and Kruh [2] and Bauer [1] give the machine descriptions we model. The recovery of the Type B machine is described by Rowlett [9] and analysed by Freeman *et al.* [3]; our Theorem 11 states the structural fact their reconstruction exploited. Modern computational cryptanalysis of the Enigma [4, 13, 7] confirms empirically what Theorem 9 states structurally: the work factor is set by the rotor state count and the plugboard is recovered afterwards, not searched. The design principle of Section 6 — that key material must move the state — is the electromechanical form of Kerckhoffs’ second requirement [6], sharpened from “the key must be changeable” to “the key must be dynamical”. Shannon’s unicity distance [11] supplies the crib-length prediction we verify. Irregular, key-controlled stepping is Hagelin’s contribution [5] and, in a different form, the SIGABA’s [10, 12]; what is new here is retaining an autonomous sub-clock *as the carrier of the period proof* while the rest of the state is message-driven.

14 Conclusion

The lesson of the two paradigms is not that reflectors and split alphabets are bad ideas. It is that a machine’s security is the size of the set of states its messages can drive it through, and that everything else — plugboards, wiring chests, patch panels — is bookkeeping unless it moves that set. The invariant taxonomy explains what the classical attacks actually consumed; the transparency theorem explains why composing the two paradigms does not help; the period-floor theorem shows that the fix does not cost the guarantee it appears to.

Coprimality does the work in both directions. It multiplies the periods of the two banks, and it makes one of them a clock that no message can perturb. A design that keeps that clock and hands the rest of the state to the message gets a proof and a diffusion rate at the same time.

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